

foundries, pulp and paper plants, chemical plants, petro-chemical industries, mining and waste management, among others.

Farm-duty motors are designed for high starting torque and moderate starting current. These are available in single- and three-phase type. Typical applications are in dairy, poultry and irrigation pumps.

DC motors

These are two-wire (power and ground) continuous-rotation motors. Range of DC motors for industrial application includes normal DC, servo and stepper motors. Nearly all DC motors used for automation are of brushless DC motor (BLDC) type.

Most of these run at a high rpm. Reduction in speed is achieved through gears (DC geared motors). BLDC motors can also be used with electronics sensitive to electrical and radio frequency interference because these do not cause a significant amount of either.

Servo motors. Servo motors are generally an assembly of four things: DC motor, gearing set, control circuit and position sensor (usually a potentiometer). These do not rotate freely, instead, angle of rotation is limited to 180 degrees (or so), back and forth. These are used where the position needs to be defined accurately, such as controlling the movement of a robotic arm or leg within a certain range.

Stepper motors. These are digital motors whose rotation is in steps defined by the step angle. Stepper motors, unlike traditional AC or DC motors, do not rotate smoothly. Each step occurs sequentially, never concurrently. Capable of rotation in both directions and precision angular incremental changes, these are more expensive than other AC or DC motors and require the use of more expensive drives, motor control systems and programming.

Applications for stepper motors include automated wire-cutting ma-

Indian Standards

Some important Indian standards on different aspects of induction motors are given below:

- IS 325 Three-phase induction motors specification
- IS 1231 Dimension of three-phase foot-mounted AC induction motors
- IS 2223 Dimensions of flange-mounted AC induction motors
- IS 4691 Degree of protection provided by enclosures for rotating electrical machinery
- IS 6362 Designation of methods for cooling of rotating electrical machines
- IS 12615 Energy-efficient induction motors
- IS 996 Single-phase AC induction motors



Fig. 2: Three-phase induction motor



Fig. 3: AC servo motor and unit



Fig. 4: Farm-duty motor

chines, robots, machine tools, precise fluid or temperature-control devices, high-speed pick-and-place machines and many other types of equipment that require very precise speed and load control.

Universal motors

These are a special type of motors designed to run on either DC or single-phase AC supply. These produce high starting torque. Most universal motors are designed to operate at higher speeds, exceeding 3500rpm.

Universal motors find their use in various home appliances like vacuum cleaners, drink and food mixers and domestic sewing machines. Higher-rated universal motors are used in portable drills, blenders, etc.

Variable frequency drive

It drives an AC electric motor by varying the frequency and voltage supplied to the electric motor. It allows speed changing of three-phase motors over a designated speed range while providing constant or variable torque as the application requires.

Selection of motor

There are two occasions when you need an induction motor. First is the replacement of existing motor and

second is *ab initio* selection of a new application.

Replacement of a motor, which has become unserviceable due to normal wear and tear, is simpler as specifications of the motor can be obtained from the plate affixed on the frame of the motor. Reasons for failure must be determined for correct selection. For example, if the motor is failing repeatedly, then obviously there is a mismatch between performance characteristics and application.

For a new application, selection can be done after evaluating horse power (HP) or kilowatt (kW) rating, speed in rpm, efficiency, duty cycle, motor frame, driving voltage and frequency, enclosures, frame size, insulation and altitude.

Horse power and speed rating.

Torque and HP are two very important characteristics that determine the size of the motor for a particular application. HP and speed ratings have to be matched to load requirements. The HP ratings are generally divided into three categories: fractional motors (0.05HP to 1HP), integral HP motors (1HP to 400HP) and large motors (100HP to 50,000HP).

Rpm of the motor should also match load requirements. Rated